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Cryptology Assignment

1. Determine the multiplicative inverse module 26 for each of the following integer

i) 1 → $26 = 1(20) + 0$

$1(1) = 1$

$1^{-1} = 1 \pmod{26}$

Multiplicative inverse of $1 = 1$

ii) 3 → $26 = 3(8) + 2$

$3 = 2(1) + 1$

$1 = 3 - 2$

$2 = 26 - 3(8)$

$1 = 3 - (26 - 3(8))$

$1 = 3 - 26 + 3(8)$

$1 = 3(4) + 26(-1)$

$3^{-1} = 9 \pmod{26}$

Multiplicative inverse of $3 = 9$

iii) 5 → $26 = 5(5) + 1$

$1 = 26 - 5(5)$

$1 = 5(5) + 26(1)$

$-5 + 26 = 21$

$5^{1} = 21 \pmod{26}$

Multiplicative inverse of $5 = 21$

iv] $7 \rightarrow 26 = 7(3)+5, \quad 5 = 26 - 7(3)$

$7 = 5(1) + 2, \quad 2 = 7 - 5$

$5 = 2(2)+1$

$1 = 5 - 2(2)$

$2 = 5 - 2(7-5)$

$1 = 5 - 2\times 7 + 2\times 5$

$1 = 3(5) - 2(7)$

$1 = 3(26 - 7(3) - 2(7)$

$1 = 3\times 36 - 11\times 7$

$1 = (-11)7 - 3\times 26, \quad -11+26 = 15$

$7^{-1} = 15 \pmod{26}$

Multiplicative inverse of $7 = 15$

v] $9 \rightarrow 26 = 9(2)+8$

$9 = 8(1)+1$

$1 = 9 - 8, \quad 8 = 26 - 9(2)$

$1 = 9 - (26 - 9(2))$

$1 = 9 + 9(2) - 26$

$1 = 3(9) - 26$

$9^{-1} = 3 \pmod{26}$

Multicative inverse of $9 = 3$

* vi] $11 \rightarrow 26 = 11(2)+4$

$11 = 4(2)+3, \quad 3 = 11 - 4(2)$

$4 = 3(1)+1, \quad 4 = 26 - 11(2)$

$1 = 4 - 3$

$1 = 4 - (11 - 4(2)) = 4 - 11 + 4(2)$

$1 = 3(4) - 11$

$1 = 3(26 - 11(2)) - 11$

$1 = 3\times 26 - 3\times(11(2)) - 11$

$1 = 3\times 26 - 7\times 11$

$1 = (-7)11 + 3 \times 26, \quad -7+26=19$

$11^{-1} = 19 \pmod{26}$

Multiplicative inverse of 11 is 19

vii] $15 \rightarrow 26 = 15(1) + 11, \quad 11 = 26 - 15$

$15 = 11(1) + 4, \quad 4 = 15 - 11(1)$

$4 = 3(1) + 1, \quad 3 = 11 - 4(2)$

$1 = 4 - 3$

$1 = 4 - (11 - 4(2))$

$1 = 3(4) - 11$

$1 = 3(15 - 11(1)) - 11$

$1 = 3 \times 15 - 4 \times 11$

$1 = 3 \times 15 - 4(26 - 15)$

$1 = 7(15) - 4(26)$

$15^{-1} = 7 \pmod{26}$

Multiplicative inverse of 11 is 19

viii] $17 \rightarrow 26 = 17(1) + 9, \quad 9 = 26 - 17(1)$

$17 = 9(1) + 8, \quad 8 = 17 - 9(1)$

$9 = 8(1) + 1$

$1 = 9 - 8$

$1 = 9 - (17 - 9(1))$

$1 = 2(9) - 12$

$1 = 2(26 - 17(1)) - 17$

$1 = 2 \times 26 - 2 \times (17(11) - 17)$

$1 = 2 \times 26 - 13 \times (17)$

$1 = (-3\,17 - 2(26)$

$17^{-1} = 23 \pmod{26}$

Multiplicative inverse of $17 = 23$

ix] 19 → $26 = 19(1) + 7$, $7 = 26 - 19$

$19 = 7(2) + 5$, $5 = 19 - 7(2)$

$7 = 5(1) + 2$, $2 = 7 - 5(1)$

$5 = 2(2) + 1$

$1 = 5 - 2(2)$

$1 = 5 - 2(7 - 5(1))$

$1 = 5 - 2 \times 7 + 2(5(1))$

$1 = 3(5) - 2(7)$

$1 = 3(19 - 7(2)) - 2(7)$

$1 = 3 \times 19 - 8(7)$

$1 = 3(19) - 8(26 - 19)$

$1 = 11(19) - 8(26)$

$19^{-1} = 11 \pmod{26}$

Multiplicative inverse of 19 = 11

x] 21 → $26 = 21(1) + 5$, $5 = 26 - 21$

$21 = 5(4) + 1$

$1 = 21 - 5(4)$

$1 = 24 - 5(26 - 21)$

$1 = 24 - (26) + 4(21)$

$1 = 5(21) - 4(26)$

$21^{-1} = 5 \bmod 26$

Multiplicative inverse of 21 = 5

xi] 23 → $26 = 23(1) + 3$, $3 = 26 - 23$

$23 = 3(7) + 2$, $2 = 23 - 3(7)$

$3 = 2(1) + 1$

$1 = 3 - 2(1)$

$1 = 3 - (23 - 3(7))$

$1 = 8(3) - 23$

$1 = 8(26 - 23) - 23$

$1 = (-9)23 + 26(8)$

$26 - 9 = 17$

$23^{-1} = 17 \bmod 26$

multiplicative inverse of 23 = 17

xii] $25 \rightarrow 26 = 25(1) + 1$

$1 = 26 - 25$

$1 = (-1)\, 25 + 26$

$11 - 1 = 25$

$25' = 25 \mod 26$

Multiplicative inverse of $25 = 25$

2. Choose any string of your choice to encrypt using the affine cipher with key values multiplier 5 & additive 8.

$R = E(K_1 * P + K_2) \mod 26$

$K_1 = 5,\ K_2 = 8$

PT		$K_1 * P$	$+K_2$	mod 26	
G	6	30	38	12	M
O	14	70	78	0	A
O	14	70	78	0	A
D	3	15	23	23	X
L	11	55	63	11	L
U	20	100	108	4	E
C	2	10	18	18	S
K	10	50	58	6	G

Cipher text = MAAXLESG

3. Encrypt "HI" using the Affine cipher with a=11 & b=5

$C = E(K * P + K_2) \mod 26$

$K_1 = 11,\ K_2 = 5$

PT		$K * P$	$+K_2$	mod 26	
H	7	77	82	4	E
I	8	88	93	15	P

Cipher text = EP